MCF-SSGN: A Multiscale Spectral–Spatial Graph Fusion Network for Hyperspectral Image Classification

Lei Wang, Zhili Zhang, Junyi Ma, Yao Ding, Yuanying Gan, Lianhui Liang, and Jie Feng

***Abstract*—Graph convolutional networks (GCNs) have been widely applied to hyperspectral image classification (HSIC). However, HSIC still faces challenges such as spectral redundancy, spatial heterogeneity, and limited labeled samples. Although recent studies have combined GCNs with convolutional neural networks (CNNs) to exploit complementary strengths, these models remain vulnerable to noise under limited supervision and often incur high computational cost. To address these issues, we propose a multiscale spectral–spatial graph fusion network (MCF-SSGN). First, a spectral-attention (SA) superpixel graph convolution submodule is introduced to perform band-aware weighted aggregation on superpixel graphs, highlighting informative spectral bands and suppressing noise for improved global representation. Second, a multiscale expert aggregation module (MEAM) is designed to dynamically generate input-dependent kernels, enhancing adaptability to diverse spatial patterns while reducing computational complexity. Finally, a dual-branch adaptive fusion mechanism is developed to learn channel-wise mixing weights for balancing and integrating the CNN and GCN branches, thereby overcoming the limitations of static fusion strategies. Extensive experiments on four widely used benchmark datasets—Indian Pines (IP), Pavia University (PU), Salinas, and WHU-Hi-HongHu—demonstrate that MCF-SSGN achieves the overall accuracies of 95.53%, 99.26%, 99.20%, and 95.65%, significantly surpassing 11 state-of-the-art baselines, including CNN-based, graph-based, Mamba-based, and hybrid models. The core code for MCF-SSGN is published at https://github.com/Majunyi310321/MCF-SSGN**

***Index Terms*—Convolutional neural networks (CNNs), graph neural networks (GNNs), hyperspectral image classification (HSIC), multiscale learning, spectral–spatial fusion.**

I. Introduction

HYPERSPECTRAL imaging (HSI) captures hundreds of contiguous and finely resolved spectral bands, enabling joint spectral–spatial analysis for precise material identification across domains such as environmental monitoring, agriculture, biomedicine, and defense applications [1], [2]. However, the increasing spectral resolution also leads to data redundancy and massive volume, posing challenges in model complexity and computational efficiency [3]. More importantly, hyperspectral image classification (HSIC) must simultaneously handle high-dimensional, non-Euclidean data distributions, limited labeled samples [4], class imbalance [5], and overfitting risks [6] while preserving fine-grained local details and capturing long-range contextual dependencies [7].

With the rise of deep learning, hierarchical feature extraction became feasible in a data-driven manner [8], [9], [10], [11]. Convolutional neural networks (CNNs), particularly 1-D and 3-D variants as well as hybrid and residual designs (e.g., 1D-CNN [12], 3D-CNN [13], SSRN [14], and HybridSN [15]), brought significant performance improvements. However, CNNs remained fundamentally constrained by their localized receptive fields. Deeper architectures and hybrid modules were introduced to extend spatial context, but often at the cost of higher computation and degraded boundary precision caused by excessive downsampling [16]. Recently, Transformer-based models were adapted for HSIC to capture long-range dependencies via self-attention mechanisms [17], [18], [19], [20]. Despite their global modeling capability, the quadratic complexity of self-attention and the reliance on regular Euclidean grids have limited their scalability and topological generalization in irregular or sparse spectral–spatial scenes. To address these issues, the newly introduced Mamba model adopts a structured state-space mechanism [21]. This architecture enables long-sequence modeling with linear time complexity. It has demonstrated potential in efficiently capturing spectral dependencies in HSI [22], [23]. However, most of the aforementioned deep learning methods are inherently designed for regular Euclidean domains, making them unsuitable for modeling complex relationships in irregular regions. Moreover, their heavy reliance on large-scale annotated data renders them prone to overfitting under limited supervision.

In contrast, graph neural networks (GNNs) can directly model non-Euclidean data structures, capturing complex pixel- or region-level relationships with reduced reliance on large-scale labeled samples. As a result, GNNs have become a promising direction to overcome the limitations of traditional

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deep learning in HSIC. Existing GNN-based methods can be broadly categorized into pure and hybrid models [24].

Pure graph-based methods focus on optimizing the graph learning architecture itself. Representative designs include graph convolutional networks (GCNs), graph recurrent neural networks (GRNNs) [25], [26], [27], [28], and graph autoencoders (GAEs). Among them, GCNs have gained prominence due to their strong feature extraction capabilities and adaptability to graph-structured data. GraphSAGE [29] was the first to introduce neighborhood sampling and aggregation into HSIC, significantly reducing spatial complexity and avoiding the high computational cost of pixel-wise graph construction. For multiscale modeling, a multiscale GraphSAGE framework was proposed in [30], which integrates context-aware mechanisms to extract features from local region graphs while mitigating noise sensitivity. Similarly, a multiscale attention-enhanced GAT model [31] was designed to simultaneously capture local and global neighborhood information, thereby improving classification accuracy and efficiency. Other methods have further refined multiscale strategies—Yin et al. [32] incorporate both pixel- and superpixel-level features to encode hierarchical semantics, while Wu et al. [33] introduce adaptive weighting across different superpixel scales to further enhance classification performance.

More recently, hybrid models have become an emerging trend. These approaches aim to combine the strengths of different architectures and are typically divided into sequential and parallel designs. Sequential models stack GNNs with other neural modules to enhance cross-domain feature representation, while parallel models employ separate branches to extract spectral, spatial, or contextual features, followed by fusion strategies to boost classification robustness. For instance, a parallel inference model proposed in [34] allows efficient handling of large-scale or cross-domain HSI without retraining. Furthermore, CEGCN [35] employs a dual-branch structure that integrates CNNs and GCNs in parallel, where CNNs capture local spatial textures on regular grids and GCNs encode global spectral–spatial dependencies in irregular regions, enabling joint modeling of pixel- and superpixel-level features. To overcome the oversmoothing issue in deep graphs, a two-branch deeper GCN (TBDGCN) was developed to synergize superpixel-level topological reasoning with pixel-level attention-based features through residual connections and DropEdge techniques [36]. In addition, Cai and Wei [37] propose a cross-attention enhanced graph convolutional model, where both horizontal and vertical attention mechanisms are used to allocate weights while leveraging GCNs to learn deep structural relationships—thus alleviating information loss caused by unidirectional attention assignment.

Despite the notable progress of hybrid GNN–CNN frameworks, several challenges remain. First, the integration of heterogeneous modules (e.g., CNNs and GNNs) often suffers from mismatched feature domains and incompatible representations, leading to suboptimal fusion. Second, many existing models rely on static fusion strategies or simple concatenation, which limits their ability to adapt to complex scene variations. Third, as multibranch designs grow in depth and complexity, computational overhead becomes a critical bottleneck, especially for large-scale or real-time applications. Therefore, a unified framework is needed that can effectively balance local–global feature extraction, reduce redundant computation, and dynamically adapt to diverse input conditions.

To address these challenges, we propose a multiscale spectral–spatial graph fusion network (MCF-SSGN) that strikes a superior balance between classification accuracy and computational efficiency. As illustrated in Fig. 1, the framework distinguishes itself through a dual-branch architecture specifically engineered to overcome the structural redundancy and high latency common in traditional hybrid models. First, instead of stacking multiple parallel convolution branches that incur prohibitive computation time, we introduce a multiscale expert aggregation module (MEAM). By dynamically generating input-dependent kernels in the parameter space, MEAM captures diverse spatial textures with significantly reduced computational cost, addressing the bottleneck of model complexity. Second, to mitigate spectral noise and feature domain mismatch, a spectral-attention superpixel GCN (SA-SGCN) is developed. It performs band-aware recalibration on a compact superpixel graph to enhance informative wavelengths before global structural modeling. Finally, to transcend the limitations of static or naïve concatenation, a bi-branch adaptive fusion module (BAFM) is proposed. By synergizing decision-level voting and feature-level integration through a learnable gated mechanism, BAFM ensures a highly nonlinear, robust alignment of heterogeneous spectral–spatial representations. This design eliminates the need for deep, redundant stacking, thereby achieving a superior equilibrium between discriminative power and inference speed across diverse and irregular hyperspectral scenes. The main contributions of this work are summarized as follows.

1) To address the challenge of jointly modeling global dependencies and local details in HSIC, we propose a dual-branch spectral–spatial framework (MCF-SSGN) that combines a superpixel-based GNN with a multiscale convolutional branch, enabling complementary feature extraction from irregular structures and regular grids.
2) To enhance spatial adaptability across heterogeneous scenes, a MEAM is introduced, which dynamically generates input-dependent convolutional kernels. This design improves the model's ability to capture multiscale spatial features and increases its flexibility in handling complex patterns.
3) To mitigate the impact of spectral redundancy and noise, we design a spectral-attentive GCN operating on a superpixel graph. It performs band-aware aggregation by reweighting informative spectral bands, thereby enhancing global structural representation and suppressing irrelevant information.
4) To avoid the limitations of static or naïve fusion, a BAFM is proposed, which learns channel-wise mixing weights to balance and integrate the two branches effectively.

The remainder of this article is organized as follows. Section II reviews the fundamental concepts of GCNs and CNNs. Section III provides a detailed description of the proposed framework. Section IV presents the experimental

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setup and results. Section V offers further discussions on model performance. Section VI outlines potential directions for future work.

II. Related Work

A. Graph CNN

GCNs enable effective feature learning on non-Euclidean domains by aggregating information from local neighborhoods. A graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with node set $\mathcal{V}$, edge set $\mathcal{E}$, and adjacency matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$. The diagonal degree matrix is given by $\mathbf{D}_i = \sum_j \mathbf{A}_{ij}$.

The normalized graph Laplacian is

$$\mathbf{L} = \mathbf{I}_n - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2} \tag{1}$$

where $\mathbf{I}_n$ is the identity matrix of size $n \times n$. Given a node feature signal x, spectral graph convolution with kernel g is defined as follows:

$$x *_G g = \mathbf{U}\left(\mathbf{U}^T x \odot \mathbf{U}^T g\right) = \mathbf{U} g \mathbf{U}^T x \tag{2}$$

where $\mathbf{U}$ is the eigenvector matrix of the Laplacian $\mathbf{L}$, $*_G$ is the graph convolution, and $\odot$ denotes the element-wise (Hadamard) product. To reduce computational cost, Defferrard et al. [38] approximate g using a truncated Chebyshev polynomial expansion

$$g_\theta(\Lambda) \approx \sum_{k=0}^{K} \theta_k T_k\left(\hat{\Lambda}\right) \tag{3}$$

where K is the order of the Chebyshev polynomial, $T_k(\cdot)$ denotes the kth Chebyshev polynomial, and θ_k is the learnable coefficient. This leads to an efficient propagation rule formulated by Kipf and Welling [39]

$$\mathbf{H}^{l+1} = \sigma\left(\tilde{\mathbf{D}}^{1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}\mathbf{H}^l\mathbf{W}^l\right) \tag{4}$$

where $\mathbf{H}^l$ denotes the node feature representations at layers $l+1$, $\mathbf{W}^l$ is the trainable weight matrix at layer l, and $\sigma(\cdot)$ is a nonlinear activation function. $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with added self-loops, and $\tilde{\mathbf{D}}$ is the corresponding degree matrix.

B. Convolutional Neural Network

CNNs are widely adopted for processing grid-structured data such as HSIs, where each pixel forms a structured node. These networks use local receptive fields and weight sharing to extract hierarchical spatial features. For an input of size $H \times W$, applying a convolutional kernel of size $k \times k$ with stride s and padding p yields an output feature map of dimensions

$$\begin{aligned} H_{\text{out}} &= \left\lfloor \frac{H + 2p - k}{s} \right\rfloor + 1 \\ W_{\text{out}} &= \left\lfloor \frac{W + 2p - k}{s} \right\rfloor + 1. \end{aligned} \tag{5}$$

Pooling layers (e.g., max or average pooling) further downsample these features, reducing resolution while preserving semantic content. However, conventional CNNs often apply fixed convolution kernels regardless of input variability, limiting adaptability in diverse scenes. To solve this problem, the expert mixing strategy generates a set of weights $\alpha_1, \alpha_2, \alpha_3$ based on the current inputs via a routing function, which realizes the dynamic change of the convolution kernel. As shown on the right side of Figs. 2 and 3, each expert—Conv1–Conv3—applies a different convolutional kernel weight $\mathbf{W}_1$, $\mathbf{W}_2$, or $\mathbf{W}_3$ to capture features at different acceptance scales. These expert outputs are then adaptively aggregated

$$\mathbf{x}_{\text{output}} = \sigma\left(\alpha_1 \cdot (\mathbf{W}_1 * \mathbf{x}) + \cdots + \alpha_n \cdot (\mathbf{W}_n * \mathbf{x})\right) \tag{6}$$

where * is a convolution operation and $\alpha_1 + \alpha_2 + \cdots + \alpha_n = 1$.

III. Proposed Methodology

The overall architecture of the proposed MCF-SSGN is illustrated in Fig. 1. It integrates five complementary modules to reduce spectral redundancy, model spatial topology, and fuse global–local features effectively. Specifically, SCNEM first compresses and enhances spectral features. SGSG then segments the image into superpixels and builds a sparse adjacency matrix to encode structural priors. Based on this graph, SA-SGCN performs band-aware aggregation to extract robust global representations. Meanwhile, MEAM applies input-conditioned convolutions with multiple receptive fields (e.g., 5×5, 7×7, and 9×9) to extract local spatial features. Finally, BAFM adaptively fuses the global and local branches using learnable channel-wise weights, yielding discriminative features for pixel-wise classification. To provide a comprehensive overview of the proposed MCF-SSGN, the overall execution flow is summarized in Algorithm 1.

A. Spectral Compression and Normalization Enhancement Module

The spectral compression and normalization enhancement module (SCNEM) serves as a lightweight preprocessing stage at the input of the MCF-SSGN framework. Its goal is to reduce spectral dimensionality and computational cost while preserving discriminative information, suppressing noise, and improving feature expressiveness. SCNEM performs global linear compression across all bands via 1×1 convolutions, followed by batch normalization (BN) and nonlinear activation (ReLU or LeakyReLU), effectively mitigating redundancy from strong interband correlations in HSI.

Structurally, SCNEM comprises three sequential 1×1 convolutional layers. Given an input feature map $\mathbf{X}^{l-1} \in \mathbb{R}^{H\times W\times B}$, the spectral compression at layer l is defined as follows:

$$\mathbf{X}_m^l(x, y) = \text{ReLU}\left(\mathbf{W}m^l \cdot \text{BN}\left(\mathbf{X}^{l-1}(x, y)\right) + B_m^l\right) \tag{7}$$

where $\mathbf{W}m^l$ and Bm^l denote the convolution weights and biases.

Let $\mathbf{X}_0$ denote the input to the SCNEM module and $\mathbf{X}$ its output. The overall compression performed by the SCNEM module can then be expressed as $\mathbf{X} = \text{SCNEM}_{L_1}(\mathbf{X}_0)$. Overall, SCNEM enhances the spectral features' discriminability and robustness while significantly reducing input complexity. The

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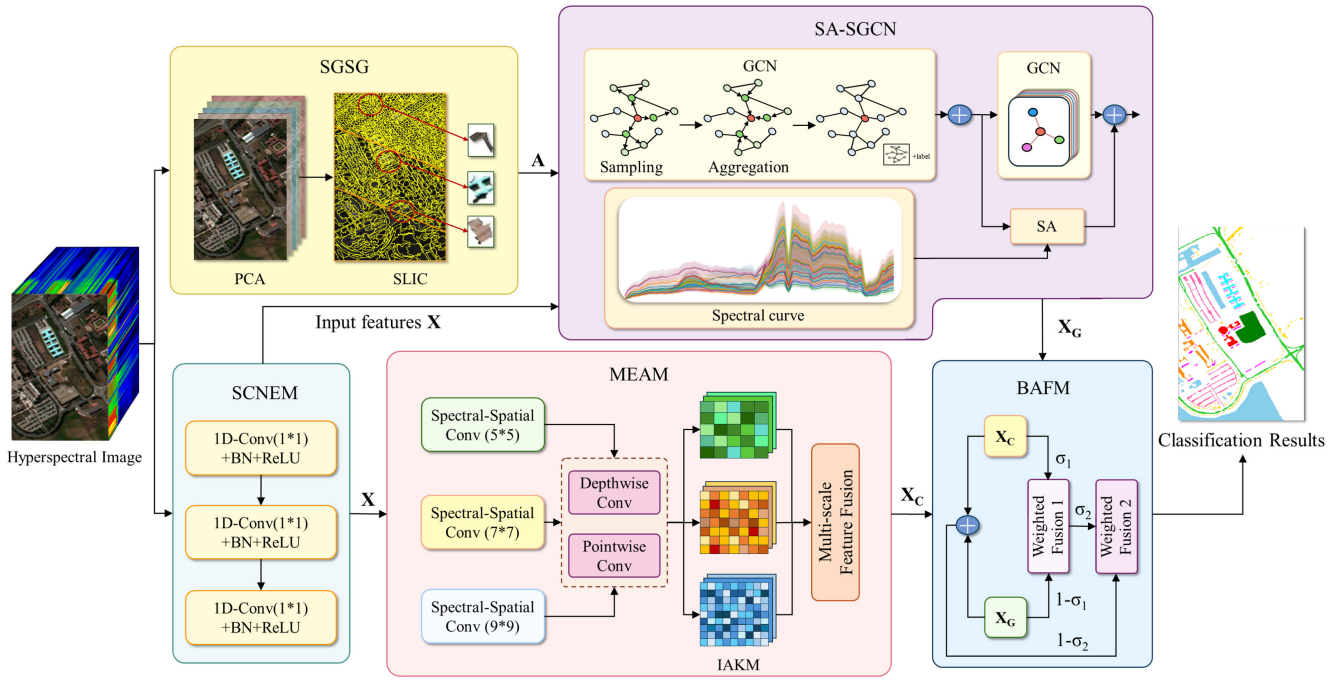


Fig. 1. Overall architecture of MCF-SSGN. The core of the network consists of four components: data processing (SCNEM and SGSGM), the SA-SGCN subnetwork, the MEAM subnetwork, and the fusion module BAFM. Arrows indicate data flow. ⊕ denotes element-wise addition. BN: batch normalization. ReLU: rectified linear unit.

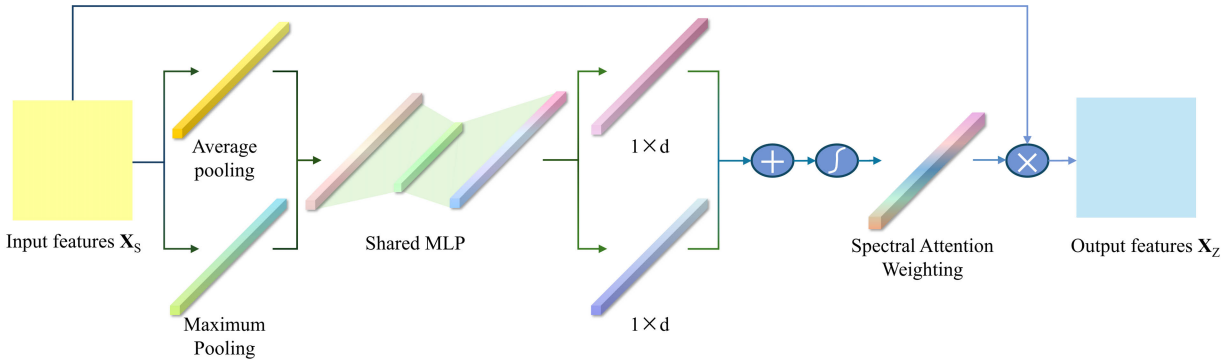


Fig. 2. Schematic of the spectral attention mechanism embedded in the SGCN branch.

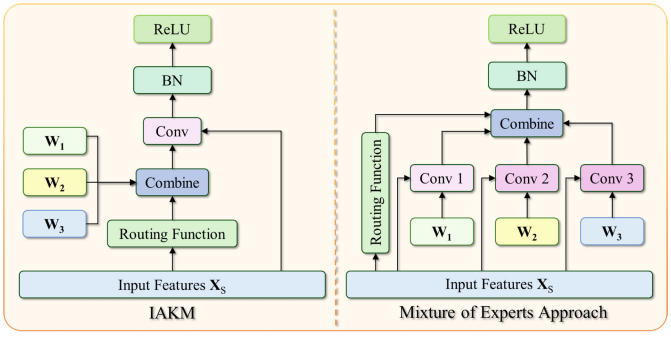


Fig. 3. Comparison between (right) traditional expert mixture convolution and (left) IAKM.

resulting representation $\mathbf{X}$ is forwarded to both the MEAM and SA-SGCN branches for joint spectral–spatial modeling.

B. Superpixel-Based Graph Structure Generation Module

To address the limitations of conventional pixel-level GCNs—namely, high computational cost, redundant nodes, and poor spatial modeling—this work introduces a superpixel-based graph structure generation module (SGSGM) at the network frontend. It aims to transform regular pixel grids into compact superpixel graphs that preserve both spatial continuity and spectral discriminability, while enabling efficient bidirectional feature mapping between pixels and graph nodes. Specifically, PCA is first applied to the spectral feature map $\mathbf{X}$ to reduce redundancy, yielding $\mathbf{X}_1 \in \mathbb{R}^{H\times W\times d}$. Then, the SLIC algorithm partitions the image into m superpixels $\mathcal{S}_1, \ldots, \mathcal{S}_m$.

A binary association matrix $\mathbf{Q} \in \mathbb{R}^{HW\times M}$ is defined to encode the pixel-to-superpixel membership

$$Q_{i,j} = \begin{cases} 1, & \text{if } \tilde{\mathbf{X}}_i \in \mathcal{S}_m \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

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Algorithm 1 Proposed MCF-SSGN Framework

Require: Input HSI data $\mathbf{X}_0 \in \mathbb{R}^{H\times W\times B}$, superpixel count m, training labels $\mathbf{Y}$.

Ensure: Classification map P_{out}.

1: **Step 1: Spectral Preprocessing (SCNEM)**
$\mathbf{X}$ = SCNEM($\mathbf{X}_0$) (linear compression and batch normalization).

2: **Step 2: Graph Structure Generation (SGSGM)**
Apply PCA to $\mathbf{X}$ to obtain the principal components $\mathbf{X}_{\text{PCA}}$;
Generate m superpixels $\{\mathcal{S}_1, \ldots, \mathcal{S}_m\}$ via SLIC on $\mathbf{X}_{\text{PCA}}$;
Construct the association matrix $\mathbf{Q}$ and the spatial adjacency matrix $\mathbf{A}^S$.

3: **Step 3: Parallel Feature Extraction**
Branch A (Graph-based):
$\mathbf{X}_S$ = Encode($\mathbf{X}$; $\mathbf{Q}$) (pixel-to-superpixel mapping);
$\mathbf{X}_H$ = SA − SGCN($\mathbf{X}_S, \mathbf{A}^S$) (spectral-attentive GCN);
$\mathbf{X}_G$ = Decode($\mathbf{X}_H$; $\mathbf{Q}$) (superpixel-to-pixel projection).
Branch B (Convolution-based):
$\mathbf{X}_C$ = MEAM($\mathbf{X}$) (multiscale expert kernel aggregation with $k \in \{5, 7, 9\}$).

4: **Step 4: Bi-branch Adaptive Fusion (BAFM)**
Compute decision-level probabilities P_C and P_G from each branch;
Concatenate aligned feature maps to obtain $\mathbf{X}_{\text{concat}}$ and its prediction P_{concat};
$P_{\text{out}} = \beta_2(\beta_1 P_C + (1-\beta_1)P_G) + (1-\beta_2)P_{\text{concat}}$, where β_1 and β_2 are learnable mixing parameters.

5: **Step 5: Training**
Update model parameters via backpropagation using Cross-Entropy loss.

6: **return:** P_{out}.

where $\tilde{\mathbf{X}}$ = Flatten($\mathbf{X}$) and Flatten is the pixel feature spread by spatial dimension. On this basis, the formula for the transformation of features between pixels and hyperpixels can be obtained as follows:

$$\mathbf{X}_S = \text{Encode}\left(\mathbf{X}; \mathbf{Q}\right) \tag{9}$$

where Encode is the graph encoder function that encodes pixel-level features $\mathbf{X}$ into superpixel-level node features $\mathbf{X}_S$ according to the $\mathbf{Q}$. Similarly, the graph decoder representation can be further obtained using $\mathbf{Q}$

$$\mathbf{X}_G = \text{Decode}\left(\mathbf{X}_S; \mathbf{Q}\right) = \text{Reshape}\left(\mathbf{Q}; \mathbf{X}_S\right) \tag{10}$$

where Decode is the graph decoder function and Reshape denotes the recovery of spatial dimension from flattened pixel features. To incorporate spatial priors, an adjacency matrix $\mathbf{A}^S$ is constructed

$$A_{i,j} = \begin{cases} 1, & \text{if } \mathcal{S}_i \text{ is spatially adjacent to } \mathcal{S}_j \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Compared to dense pixel-level graphs, SGSGM significantly reduces node count and computational burden while preserving essential spatial boundaries. The resulting node features $\mathbf{X}_S$ and adjacency matrix $\mathbf{A}^S$ are passed to the SA-SGCN for structure-aware spectral–spatial learning.

C. SA Superpixel Graph Convolution Sub Module

Following the front-end preprocessing and spectral compression, although global redundancy is largely mitigated, HSIs still exhibit intricate local spectral–spatial patterns. These local regions often present strong spatial continuity while differing significantly in their spectral signatures. Conventional GCNs with fixed-graph topologies struggle to capture such coupled dependencies. To address this, we design an SA-SGCN, which builds upon the outputs of SGSGM and enhances both spectral selectivity and adaptive spatial structure learning via attention and dynamic graph mechanisms [20].

As illustrated in Fig. 2, a spectral attention mechanism is applied to each layer to dynamically recalibrate channel-wise responses. Unlike conventional CNNs that process regular grids, SA-SGCN operates on a non-Euclidean domain. Therefore, the spectral attention mechanism is reformulated to capture the global spectral context by aggregating information across all superpixel nodes. Given the input node features $\mathbf{X}_S^{l-1} \in \mathbb{R}^{M\times d}$ from layer $l-1$ (where M denotes the number of superpixel nodes), global average pooling and max pooling are performed across the node set to extract two spectral descriptors

$$F_{\text{avg}}^{\text{d}} = \frac{1}{M}\sum_{i=1}^{M} \mathbf{x}_{S,i} \tag{12}$$

$$F_{\text{max}}^{d} = \max_{i\in\{1,\ldots,M\}} \left(\mathbf{x}_{S,i}\right) \tag{13}$$

where $\mathbf{x}_{S,i}$ represents the d-dimensional feature vector of the ith superpixel node.

These are passed through a shared MLP with learnable weights W_0 and W_1, producing attention weights $\alpha^l \in \mathbb{R}^{1\times 1\times d}$

$$\alpha^l = \sigma\left(W_1\left(W_0\left(F_{\text{avg}}^{d}\right)\right) + W_1\left(W_0\left(F_{\text{max}}^{d}\right)\right)\right) \tag{14}$$

where $\sigma(\cdot)$ denotes a nonlinear activation (e.g., sigmoid). The resulting attention vector is applied to modulate the original node features

$$\mathbf{X}_Z^l = \alpha^l \otimes \mathbf{X}_S^{l-1} \tag{15}$$

where $\otimes$ denotes channel-wise multiplication. This adaptive weighting emphasizes informative bands while suppressing redundant or noisy spectral components, enhancing spectral discriminability before graph aggregation.

In the spatial domain, SA-SGCN generates a layer-wise dynamic adjacency matrix to overcome the limitations of fixed topologies. It computes pairwise feature similarity using learnable projections W_θ and W_ϕ, modulated by the prior adjacency mask $\mathbf{A}^S$ and reinforced with self-loops

$$\mathbf{A}^l = \sigma\left(\left(\mathbf{X}_S^{l-1}W_\theta\right)\left(\mathbf{X}_S^{l-1}W_\phi\right)^{\text{T}} \odot \mathbf{A}^S\right) + \mathbf{I} \tag{16}$$

where $\odot$ denotes element-wise multiplication and $\mathbf{I}$ ensures self-connections. This design preserves spatial adjacency priors while allowing data-driven edge weighting that adapts to local feature distributions.

The final graph convolution operation at layer l is formulated as follows:

$$\mathbf{X}_H^l = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{A}^l\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{X}_S^{l-1}\mathbf{W}^l\right) + \mathbf{X}_Z^l \tag{17}$$

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where $\tilde{\mathbf{D}}$ is the degree matrix of $\mathbf{A}^l$, $\mathbf{W}^l$ is the layer-specific weight matrix, and $\mathbf{X}_Z^l$ provides residual spectral attention features.

By integrating spectral recalibration and adaptive topology learning, SA-SGCN effectively models both global and local dependencies with reduced computational overhead. Compared to traditional pixel-level GCNs or fixed-graph methods, this approach enhances robustness, improves class separability, and yields richer spatial–spectral representations for subsequent classification branches.

D. Multiscale Expert Aggregation Module

Following the spectral compression and normalization enhancement performed by SCNEM, the network receives high-quality spectral embeddings for further spatial–spectral modeling. However, using fixed convolution kernels often struggles to maintain robust performance under varying spatial scales and complex land-cover distributions. To address this, we propose a MEAM that leverages input-adaptive kernel modulation (IAKM) to dynamically generate convolutional kernels [40], enabling fine-grained multiscale texture modeling and adaptive feature extraction.

Compared with traditional mixture-of-experts (MoE) or conditional convolutions that aggregate the outputs of multiple parallel convolution branches, IAKM operates directly in the kernel parameter space. Specifically, for HSI data with high spectral dimensionality, traditional MoE incurs redundant computational overhead by executing multiple heavy convolution operations. In contrast, IAKM first generates a single, input-conditioned kernel $\mathbf{W}_{\text{IAKM}}$ by combining multiple expert kernels and then applies it to the feature map in a single pass. This mechanism not only dramatically reduces the computation time and memory footprint but also enables the model to adaptively synthesize kernels that are optimized for specific spatial–spectral structures (e.g., fine edges versus homogeneous regions), making it more suitable for complex and heterogeneous hyperspectral scenes.

As illustrated in Fig. 1, MEAM consists of three parallel branches, each comprising a $k \times k$ depthwise convolution layer followed by a 1×1 pointwise fusion. Both layers adopt the IAKM strategy. As shown on the left side of Fig. 3, IAKM, compared to the traditional expert mixture convolution on the right side, does not rely on static filters, but dynamically combines convolution weights from a pool of expert kernels according to the input-related routing coefficients, so that the enhancement of the model capacity is no longer strongly correlated with the size of the input space,. Thus, higher computational efficiency is realized. This mechanism can be expressed as follows:

$$\mathbf{W}_{\text{IAKM}} = \sum_{m=1}^{M} \alpha_{k,m} \mathbf{W}_{k,m} \tag{18}$$

where $\mathbf{W}_{k,m}$ denotes the mth expert kernel and $\alpha_{k,m}$ is the input-conditioned routing weight. The routing vector $\boldsymbol{\alpha}_k$ is obtained as follows:

$$\boldsymbol{\alpha}_k = f_{\text{route}}\left(\text{GAP}\left(\mathbf{X}\right) \cdot \mathbf{R}\right) \tag{19}$$

where GAP$(\cdot)$ denotes global average pooling and $\mathbf{R}$ is a learnable projection matrix that maps pooled features to expert-specific weights. This allows MEAM to modulate convolution operations based on global input context while maintaining low computational overhead—since only a single weighted kernel needs to be evaluated per spatial location. The input-aware convolution is applied using the dynamically generated kernel

$$\mathbf{X}_{\text{depth}}^k = \sigma\left(\text{BN}\left(\mathbf{X} \star_{\text{depth}} \mathbf{W}_{\text{IAKM}}\right)\right) \tag{20}$$

where $\star_{\text{depth}}$ denotes depthwise convolution and $\sigma(\cdot)$ is the activation function. Subsequently, a 1×1 IAKM-based pointwise convolution aggregates channel-wise information

$$\mathbf{X}_{\text{out}}^k = \sigma\left(\text{BN}\left(\mathbf{X}_{\text{depth}}^k * \mathbf{W}_{\text{out}}\right)\right) \tag{21}$$

where $\mathbf{W}_{\text{out}}$ is the fusion kernel. This two-step process ensures both spatial adaptation and channel-wise integration without altering spatial resolution.

To achieve scale-aware modeling, MEAM adopts three branches with kernel sizes $k \in \{5, 7, 9\}$. The 5×5 branch captures fine-grained edges and small structures, the 7×7 branch targets medium-scale features such as vegetation patches or road boundaries, and the 9×9 branch extracts low-frequency patterns over large homogeneous regions. The outputs from all branches are concatenated along the channel dimension

$$\mathbf{X}_C = \text{Concat}\left(\mathbf{X}_{\text{out}}^5, \mathbf{X}_{\text{out}}^7, \mathbf{X}_{\text{out}}^9\right) \tag{22}$$

which preserves scale-specific details while enabling cross-scale feature interaction in subsequent layers. Compared to conventional fixed-kernel designs, MEAM offers dynamic kernel modulation and efficient multiscale aggregation, making it particularly effective in regions with complex boundaries, heterogeneous classes, or significant object scale variation. The resulting feature map $\mathbf{X}_C$ serves as input to the downstream fusion module, forming a robust foundation for spatial–spectral joint optimization.

E. Bi-Branch Adaptive Fusion Module

After performing local texture modeling via the MEAM convolutional branch and global structural modeling via the SA-SGCN graph branch, a critical challenge lies in effectively integrating their complementary features. To this end, we design a BAFM, which facilitates precise alignment and optimal fusion of heterogeneous features through four stages: dimensionality reduction, spatial mapping, channel fusion, and adaptive weighting. This integration enhances the pixel-wise discriminative capability of the overall framework.

Although the MEAM branch produces multiscale features $\mathbf{X}_C \in \mathbb{R}^{H \times W \times 192}$ rich in spatial textures, its high channel dimensionality (192) leads to increased computation and potential overfitting due to redundancy. Therefore, we apply a 1×1 convolution for linear projection and dimensional compression, reducing it to 64 channels while preserving essential multiscale representations and suppressing noise.

Simultaneously, the output of the SA-SGCN graph branch, denoted as $\mathbf{X}_H$, exists in superpixel space and requires alignment with pixel space. To achieve this, we utilize the spatial

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graph mapping matrix $\mathbf{Q}$, generated by the SPGG module, to project superpixel features back to pixel space. This ensures that $\mathbf{X}_H$ shares the same spatial resolution as $\mathbf{X}_C$ and enables subsequent fusion.

Simultaneously, the output of the SA-SGCN graph branch, denoted as $\mathbf{X}_H \in \mathbb{R}^{M\times d}$, exists in the non-Euclidean superpixel node space. To enable fusion with the CNN branch, we align $\mathbf{X}_H$ with the regular pixel grid using the association matrix $\mathbf{Q} \in \mathbb{R}^{HW\times M}$. Specifically, the aligned graph feature map $\mathbf{X}_G \in \mathbb{R}^{H\times W\times d}$ is reconstructed by projecting the superpixel features back to the pixel space according to (10). This operation ensures that $\mathbf{X}_G$ and the CNN feature map $\mathbf{X}_C$ share identical spatial resolution and coordinate systems, facilitating subsequent integration.

Based on spatial alignment, instead of directly applying linear combinations on the high-dimensional features (which could mathematically degenerate into a single linear projection), our BAFM employs a **dual-level (decision and feature) fusion strategy** coupled with nonlinear activations.

First, at the **decision level**, the spatially aligned features from the two branches ($\mathbf{X}_C$ and $\mathbf{X}_G$) are independently mapped into class probability distributions using their respective linear classifiers followed by a Softmax activation

$$P_C = \text{Softmax}\left(\mathbf{W}_C\mathbf{X}_C + \mathbf{b}_C\right) \tag{23}$$

$$P_G = \text{Softmax}\left(\mathbf{W}_G\mathbf{X}_G + \mathbf{b}_G\right) \tag{24}$$

where $\mathbf{W}_C$ and $\mathbf{W}_G$ are the weights of the branch-specific classifiers and $\mathbf{b}_C$ and $\mathbf{b}_G$ are the bias terms. We then apply an adaptive weighted voting mechanism to integrate these independent predictions

$$P_{\text{vote}} = \beta_1 P_C + (1-\beta_1) P_G \tag{25}$$

where $\beta_1 \in [0,1]$ is a learnable scalar dynamically balancing the decision-level contributions of the CNN and GCN branches.

Simultaneously, at the **feature level**, we explicitly concatenate the aligned features to capture cross-domain interactions. This concatenated representation is then projected into the prediction space

$$\mathbf{X}_{\text{concat}} = \text{Concat}\left(\mathbf{X}_C, \mathbf{X}_G\right) \tag{26}$$

$$P_{\text{concat}} = \text{Softmax}\left(\mathbf{W}_{\text{concat}}\mathbf{X}_{\text{concat}} + \mathbf{b}_{\text{concat}}\right). \tag{27}$$

Finally, to synergize the robustness of decision-level voting and the representational richness of feature-level concatenation, a second-stage adaptive gate β_2 is introduced to yield the ultimate prediction

$$P_{\text{out}} = \beta_2 P_{\text{vote}} + (1-\beta_2) P_{\text{concat}}. \tag{28}$$

To ensure that the fusion weights are strictly bounded and smoothly optimizable, they are parameterized via the sigmoid function as $\beta_1 = \sigma(\gamma_1)$ and $\beta_2 = \sigma(\gamma_2)$, where γ_1 and γ_2 are learnable scalar parameters optimized via backpropagation. By introducing intermediate independent classifiers and nonlinear Softmax transformations, this hierarchical design ensures that the fusion process is highly nonlinear and cannot be trivially reduced to a single linear combination, thereby significantly enhancing the discriminative capacity of the network.

IV. Experiments and Results

Extensive experiments were carried out on four widely used hyperspectral datasets: Indian Pines (IP), Pavia University (PU), Salinas (SA), and WHU-Hi-HongHu (HongHu). All models were implemented in PyTorch and trained on a workstation equipped with an Intel Xeon Platinum 8151 CPU @ 3.40 GHz and an NVIDIA Quadro GV100 GPU. Unless otherwise specified, training was performed using the Adam optimizer with a cross-entropy loss function, a fixed learning rate of 3×10^{-3}, and 300 epochs.

A. Experimental Settings

1) Evaluation Metrics: Three standard metrics were employed: overall accuracy (OA), average accuracy (AA), and Cohen's kappa coefficient (κ). OA measures the proportion of correctly classified pixels over the entire test set, serving as an indicator of overall performance. AA computes the mean accuracy across all classes, thus alleviating the impact of class imbalance. The κ coefficient quantifies the agreement between predicted and ground truth labels while accounting for random chance, offering a more robust measure than OA under imbalanced distributions. In addition to these quantitative metrics, qualitative evaluation was conducted via visual inspection of classification maps, emphasizing spatial coherence and boundary sharpness. Furthermore, computational efficiency was assessed in terms of model size (number of parameters), training time, and inference latency—providing a balanced perspective between predictive performance and deployment cost.

2) Comparison Methods: To rigorously assess the performance of the proposed MCF-SSGN model, 11 representative classification methods were selected, spanning a broad spectrum of technical paradigms and development stages. The comparison includes one classical machine learning method: support vector machine (SVM) [41]; two conventional deep learning models: 3-D CNN (3D-CNN) [13] and spectral–spatial residual network (SSRN) [14]; two advanced Transformer-based models: SpectralFormer (SF) [42] and multigranularity hybrid transformer (MGHT) [43]; four state-of-the-art graph-based approaches: convolution-enhanced GCN (CEGCN) [35], attention-based multihop graph and multiscale convolutional fusion network (AMGCFN) [44], CNN–GAT-based weighted feature fusion network (WFCG) [45], and graph convolution–transformer fusion network (GTFN) [20]; as well as two recent Mamba-based architectures: the spatial–spectral selective structured state-space model (S^2Mamba) [21] and HSI-MFormer [22], which integrates Mamba with Transformer experts for hyperspectral representation learning. This diverse and extensive set of baselines provides a robust comparative framework for evaluating MCF-SSGN across multiple modeling paradigms, from traditional classifiers to cutting-edge sequential and graph-based architectures.

3) Sample Selection: All methods were evaluated using a consistent data partitioning protocol. Thirty randomly selected labeled samples from each class were used for training (15 samples were selected if a class did not satisfy

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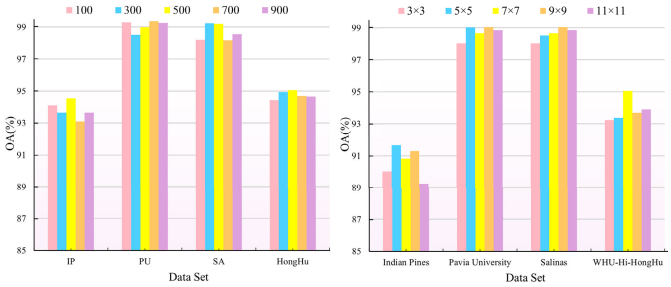


Fig. 4. Impact of (left) superpixel granularity and (right) MEAM kernel size on OA across datasets.

30 samples), and another 30 samples (including at least one sample from each class) were used for validation. The remaining labeled pixels were reserved for testing. This setup reflects realistic scenarios with limited annotated data for training and validation, while leveraging the majority of labels to evaluate generalization performance. All reported results were obtained by averaging the outcomes from ten independent runs, each with different randomly selected training and validation sets.

B. Key Hyperparameter Settings

To thoroughly evaluate the impact of multiscale feature extraction, graph construction, and cross-branch fusion in MCF-SSGN, four key hyperparameter groups were explored: 1) superpixel granularity in SGSGM, controlled by the SLIC segmentation scale λ (i.e., the target number of superpixels); 2) kernel size $k \times k$ in MEAM; 3) fusion weights β_1 and β_2 in BAFM; and 4) network depth l of the core submodules. Systematic ablation studies were performed to analyze how these parameters affect model performance and to offer practical insights for hyperparameter tuning.

1) Effect of Segmentation Scale: To evaluate the effect of superpixel granularity in SGSGM, we varied the SLIC segmentation scale across five levels: 100, 300, 500, 700, and 900 superpixels. As shown in Fig. 4 (left), this parameter governs the tradeoff between boundary fidelity and intraregion compactness. Smaller scales risk oversegmentation and noise sensitivity, whereas overly coarse granularity may smooth class boundaries and obscure small structures. Experimental results reveal dataset-dependent sensitivity. For PU and SA, OA remained consistently high (near 99%) across all scales, indicating strong robustness. In contrast, IP exhibited a clear performance peak at 500 superpixels (94.5%), with noticeable drops when the scale decreased to 300 or increased to 700. HongHu displayed moderate sensitivity, with optimal OA at 300–500 superpixels and slight degradation at extreme scales. These patterns align with scene complexity: PU/SA contain large, high-contrast regions, while IP exhibits fine-grained, spectrally similar classes that benefit from mid-scale superpixel partitioning. For all subsequent experiments, optimal SLIC counts were adopted per dataset: 500 (IP), 700 (PU), 300 (SA), and 500 (HongHu).

2) Effect of Convolutional Kernel Size: To assess the spatial–spectral modeling capability of the MEAM, we varied the convolutional kernel size $k \times k$ within $\{3, 5, 7, 9, 11\}$ under a single-layer configuration (Fig. 4, right). Overall, mid-sized kernels (5×5, 7×7, and 9×9) consistently yielded competitive results across datasets, balancing local detail extraction and global context modeling. On IP, large kernels (9×9 and 11×11) tended to oversmooth small field patches, reducing class separability. In contrast, PU and SA, dominated by larger urban or crop regions, benefited from wider context windows, where 7×7 or 9×9 offered optimal results. Notably, 3×3 was consistently suboptimal due to limited receptive field and insufficient context capture. In the HongHu scene, 7×7 emerged as the best-performing kernel, striking a favorable balance between boundary sharpness and contextual awareness. Therefore, the kernel sizes $\{5, 7, 9\}$ were adopted as default configurations in the final model due to their strong and stable performance.

3) Effect of Network Depth: The impact of network depth was evaluated by varying the number of layers $l \in \{1, 2, 3, 4\}$ in both SA-SGCN and CC-MSSE modules. As shown in Fig. 5, optimal performance was generally achieved when both branches were set to two or three layers. Shallower configurations ($l = 1$) resulted in insufficient feature abstraction, while deeper ones ($l = 4$) tended to overfit, especially under limited supervision. On IP and PU, performance peaked at moderate depth, whereas on SA and HongHu, accuracy improved steadily with depth up to a saturation point. Based on these observations, a default depth of $l = 2$ was adopted in all main experiments to balance accuracy and computational efficiency.

4) Effect of Fusion Weights: The impact of the dual-level fusion mechanism in BAFM was assessed by temporarily replacing the dynamically learned parameters γ_1 and γ_2 with constant scalars. As shown in Fig. 6, optimal performance was generally observed near $(\gamma_1, \gamma_2) \in [0, 0.1]$, with the highest OA achieved at or around $(0.0, 0.0)$ for SA. These results indicate that balanced contributions from both the independent branch probabilities (decision-level) and the concatenated representations (feature-level) are essential for robust generalization. On IP, performance declined as γ_2 increased, suggesting that spectral–graph cues were more critical under subtle interclass variations. In contrast, SA and HongHu exhibited greater tolerance, with stable accuracy across a broader range of fusion weights, likely due to richer spatial textures. Overall, this balanced nonlinear fusion was found to enhance both discriminative ability and robustness, while disproportionate weighting led to degradation. Unless otherwise stated, $(\gamma_1, \gamma_2) = (0.0, 0.1)$.

C. Classification Results and Analysis

1) Comparative Experiment Analysis: The classification performance of MCF-SSGN is summarized in Tables I–IV, where the reported results represent the mean classification accuracy and the corresponding standard deviation obtained from ten independent experimental runs, demonstrating consistent superiority across all datasets. On IP, MCF-SSGN achieves an OA of 95.53%, outperforming AMGCFN and S^2Mamba by 2.13% and 9.05%, respectively, with notable improvements in Classes 3 and 8. On PU, it reaches 99.26% OA, showing high reliability in structured urban categories such as Classes 3 and 5. For SA, MCF-SSGN obtains 99.20%

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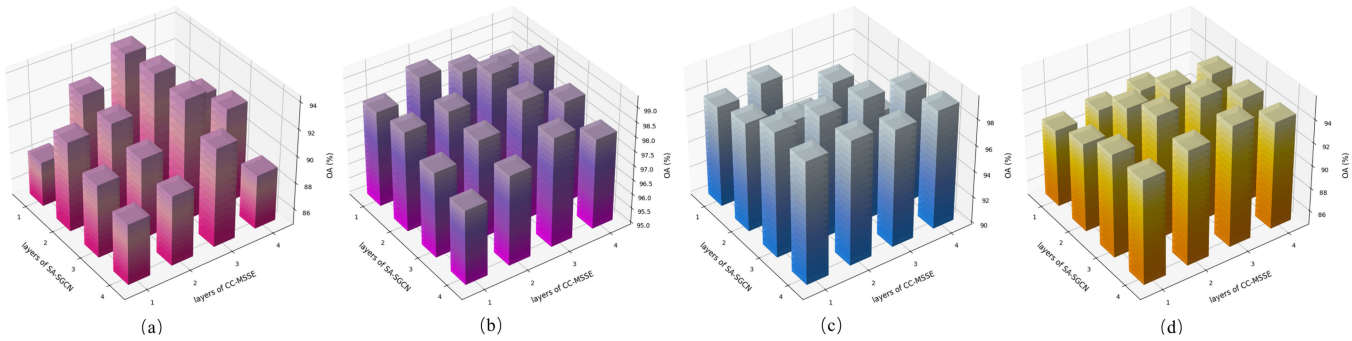


Fig. 5. Impact of network depth on OA. (a)–(d) IP, PU, SA, and HongHu datasets, respectively.

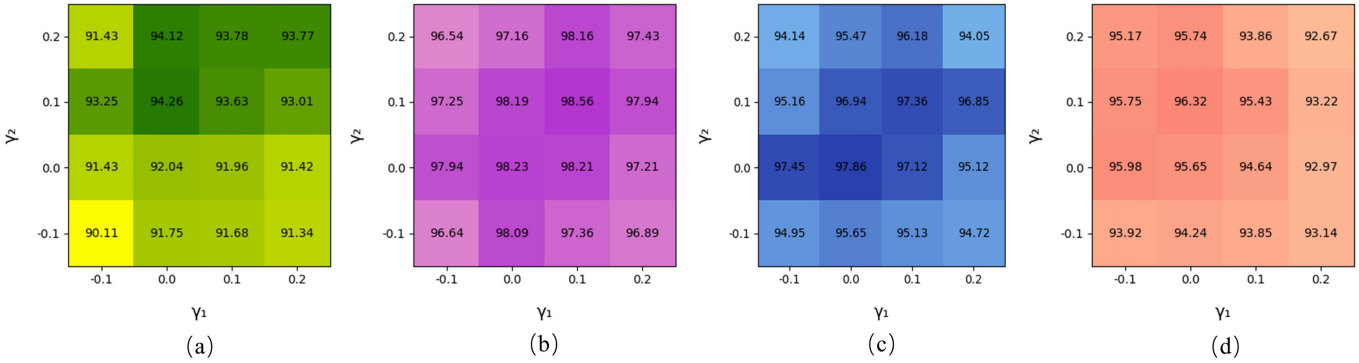


Fig. 6. Effect of fusion weights on OA. (a)–(d) IP, PU, SA, and HongHu datasets, respectively.

TABLE I

COMPARISON ON THE IP DATASET. THE BEST RESULTS ARE IN BOLD

Class	SVM	3D-CNN	SSRN	SF	MGHT	CEGCN	AMGCFN	WFCG	GTFN	S^{2}Mamba	HSI-MFormer	MCF-SSGN
1	39.04±1.88	77.83±7.89	92.12±6.50	90.65±3.23	90.63±2.55	93.75±0.54	85.71±1.44	93.16±1.95	96.77±2.72	76.37±1.34	**100.00±0.00**	96.66±1.57
2	61.02±6.51	83.61±4.75	86.63±8.00	67.41±7.92	81.34±0.63	84.52±4.62	84.07±2.58	87.73±2.32	89.20±1.55	**92.09±1.12**	87.54±0.76	92.04±4.63
3	71.68±3.15	87.59±2.02	75.55±9.25	80.75±3.56	92.00±1.71	83.03±5.98	**99.52±0.38**	96.25±0.31	81.50±3.46	**100.00±0.00**	96.27±0.85	99.12±0.42
4	30.15±7.68	79.67±6.41	81.02±4.11	96.59±2.59	92.45±1.47	84.66±1.24	79.20±1.35	86.67±0.16	99.52±0.22	91.28±0.95	**100.00±0.00**	**100.00±0.00**
5	84.16±4.70	79.31±4.81	89.97±4.55	**95.67±1.48**	92.70±1.80	91.57±2.05	94.58±2.83	85.84±4.57	94.26±1.08	99.86±0.17	95.65±0.91	91.13±2.53
6	90.09±1.90	89.08±5.01	92.90±3.21	92.58±3.95	91.91±2.34	99.71±0.13	97.03±1.45	92.46±0.15	96.14±3.17	**100.00±0.00**	97.67±0.63	99.13±0.47
7	52.65±3.92	76.04±7.64	94.30±4.21	90.86±2.17	88.34±3.29	99.23±0.22	81.23±1.75	92.38±4.31	92.31±1.28	79.74±1.53	**100.00±0.00**	**100.00±0.00**
8	89.30±2.37	79.92±7.62	95.07±3.76	95.74±1.24	94.00±0.68	99.55±0.34	99.12±0.09	83.24±0.12	98.88±1.21	76.22±1.66	**100.00±0.00**	**100.00±0.00**
9	39.42±8.32	68.65±1.60	99.25±0.34	92.99±1.01	94.44±0.47	98.79±0.41	**100.00±0.00**	95.24±3.69	**100.00±0.00**	88.48±1.07	**100.00±0.00**	**100.00±0.00**
10	62.10±4.84	85.65±5.79	83.69±7.76	63.03±7.22	85.92±4.16	84.23±1.62	87.68±2.94	92.73±0.33	93.63±2.06	**100.00±0.00**	92.70±0.98	95.32±0.60
11	79.94±4.78	87.16±5.01	90.53±4.22	77.17±5.82	91.62±1.43	96.82±0.66	97.62±2.06	**98.47±0.31**	81.65±2.24	96.06±0.92	87.17±1.00	93.05±1.96
12	59.97±1.42	81.36±4.04	89.97±4.71	91.22±2.82	87.70±3.47	89.34±2.95	77.28±3.74	86.11±0.18	84.90±1.96	**98.79±0.85**	93.42±0.76	94.15±2.44
13	90.05±3.92	78.91±6.28	98.62±1.25	93.12±1.43	93.22±0.68	98.46±0.75	93.26±0.93	99.68±0.09	98.86±0.33	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
14	89.81±3.25	79.73±7.96	99.16±0.12	92.82±1.16	93.33±0.74	98.91±0.97	99.67±0.18	84.87±0.13	99.03±0.48	**100.00±0.00**	99.05±0.49	99.59±0.66
15	48.73±1.02	72.79±6.97	88.64±5.14	77.02±6.12	92.78±5.09	99.43±0.37	**99.73±0.21**	92.59±0.69	97.47±1.76	**100.00±0.00**	**100.00±0.00**	98.02±0.48
16	80.13±1.65	80.01±4.26	98.86±1.07	95.94±1.04	90.55±4.42	98.38±0.55	97.80±2.09	89.57±5.68	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA (%)	73.97±2.83	76.25±3.13	87.53±3.32	81.54±2.56	91.32±1.86	92.51±2.35	93.40±3.94	91.74±2.71	90.09±3.26	86.48±1.28	93.33±0.85	**95.53±1.18**
AA (%)	66.82±1.69	76.71±5.52	89.12±4.98	87.10±3.50	91.43±1.99	94.49±1.41	92.02±1.19	90.49±1.97	94.01±2.28	93.68±1.02	96.84±0.71	**97.22±0.37**
Kappa	70.29±2.98	76.44±4.88	85.76±3.36	79.02±3.15	90.10±2.13	91.36±2.25	92.47±3.26	90.58±2.79	88.73±3.30	84.61±1.41	93.74±0.66	**94.88±1.34**

TABLE II

COMPARISON ON THE PU DATASET. THE BEST RESULTS ARE IN BOLD

Class	SVM	3D-CNN	SSRN	SF	MGHT	CEGCN	AMGCFN	WFCG	GTFN	S^{2}Mamba	HSI-MFormer	MCF-SSGN
1	83.82±1.56	84.01±0.65	91.08±2.49	66.22±6.89	97.40±0.14	**99.24±0.18**	88.19±4.57	91.61±3.62	84.21±0.73	86.90±0.98	85.89±0.82	98.05±0.35
2	94.75±0.24	83.32±2.75	**100.00±0.00**	78.06±5.43	97.93±0.06	97.50±2.61	98.92±0.80	**100.00±0.00**	95.05±0.29	96.17±0.55	97.89±0.45	99.76±0.16
3	56.59±4.22	81.75±1.56	80.46±3.26	93.13±2.35	82.09±0.17	72.67±5.51	70.79±5.89	92.16±1.28	78.44±0.79	95.65±0.76	95.88±0.67	**99.96±0.04**
4	74.34±0.96	84.62±2.68	86.33±2.69	53.75±8.12	81.62±0.28	88.44±2.65	95.60±2.46	72.08±6.44	89.72±0.83	**98.12±0.34**	95.42±0.74	97.41±0.97
5	72.62±0.65	81.45±4.96	94.87±1.23	98.70±0.46	99.83±0.13	96.16±0.19	99.55±0.28	99.69±0.09	98.94±0.46	**100.00±0.00**	99.82±0.09	**100.00±0.00**
6	71.06±1.63	75.46±9.68	82.45±0.45	97.43±1.23	78.81±0.19	**99.56±0.17**	98.94±0.97	97.96±1.35	99.12±0.23	97.24±0.92	99.93±0.18	99.49±0.50
7	68.03±2.58	76.41±1.14	95.76±1.20	85.69±1.49	94.97±0.10	93.76±4.72	92.79±1.22	97.02±1.24	97.62±1.89	98.00±0.63	99.97±0.11	**100.00±0.00**
8	81.87±1.96	84.21±0.88	90.06±0.96	86.11±3.57	94.84±0.39	91.09±3.19	99.01±0.30	88.29±6.24	78.75±0.79	95.04±1.09	84.23±1.03	**99.44±0.23**
9	68.99±3.20	82.20±0.42	81.29±3.69	36.75±12.34	**99.71±0.22**	94.91±1.25	69.17±7.25	99.89±0.08	96.95±1.36	99.67±0.42	97.89±0.53	98.28±0.28
OA (%)	83.73±0.94	82.29±1.97	93.05±1.79	78.17±5.34	93.40±0.81	95.40±2.09	94.83±2.98	94.97±2.19	91.50±0.37	95.12±0.48	94.94±0.47	**99.26±0.05**
AA (%)	74.67±1.16	81.49±2.02	89.14±1.27	77.32±4.12	91.91±0.86	92.59±2.12	90.35±1.35	93.19±1.57	90.98±0.76	96.31±0.69	95.21±0.32	**99.16±0.10**
Kappa	78.44±0.79	76.54±1.66	90.82±1.49	72.27±4.53	91.25±1.12	93.84±7.28	93.17±2.20	93.33±2.76	88.81±0.43	93.58±0.59	93.32±0.62	**99.02±0.07**

OA and maintains over 98% accuracy on 14 out of 16 classes, including significant gains in Classes 5 and 8. On the complex HongHu scene, it delivers 95.65% OA, surpassing HSI-MFormer by 3.98% and improving the accuracy of

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TABLE III

COMPARISON ON THE SA DATASET. THE BEST RESULTS ARE IN BOLD

Class	SVM	3D-CNN	SSRN	SF	MGHT	CEGCN	AMGCFN	WFCG	GTFN	S^{2}Mamba	HSI-MFormer	MCF-SSGN
1	98.94±1.01	97.79±0.13	99.51±0.34	98.56±0.95	99.80±1.01	96.15±1.72	95.03±0.88	98.62±0.65	**100.00±0.00**	87.68±1.20	99.98±0.04	**100.00±0.00**
2	92.33±1.70	94.55±1.72	99.54±0.35	94.62±0.11	99.76±0.13	99.96±0.06	**100.00±0.00**	**100.00±0.00**	99.95±0.02	92.42±1.01	**100.00±0.00**	**100.00±0.00**
3	94.75±0.42	92.48±1.21	99.63±0.16	98.47±1.36	99.85±0.17	99.93±0.05	97.56±0.79	**100.00±0.00**	**100.00±0.00**	65.31±1.53	**100.00±0.00**	**100.00±0.00**
4	98.91±0.31	98.63±0.14	99.53±0.25	98.33±0.99	99.01±0.44	**100.00±0.00**	82.72±3.28	99.92±0.06	99.49±0.09	98.90±0.45	99.61±0.06	99.95±0.02
5	96.27±0.93	90.08±0.73	96.04±3.26	97.65±1.98	98.31±1.07	97.41±1.72	97.11±0.71	99.01±0.54	93.35±3.56	92.56±1.12	96.82±0.03	**99.39±0.23**
6	97.49±0.58	96.73±1.27	99.48±0.32	99.43±0.21	99.32±0.06	99.39±0.60	97.51±0.10	98.99±0.50	98.77±2.00	99.90±0.33	**99.96±0.03**	**100.00±0.00**
7	95.22±0.43	97.74±0.14	99.35±0.55	98.29±1.53	96.81±0.37	99.86±0.15	98.45±0.64	**99.97±0.02**	99.89±0.10	50.41±1.26	99.77±0.02	99.99±0.01
8	65.59±1.75	72.31±0.59	92.24±5.36	75.12±8.33	88.45±1.48	90.23±1.77	95.13±0.85	88.36±4.22	85.26±5.78	58.62±1.13	93.57±0.07	**98.80±0.19**
9	83.96±1.34	95.23±0.86	97.79±0.37	94.68±3.64	98.23±0.58	95.25±0.31	**100.00±0.00**	95.59±0.45	99.45±0.37	99.60±0.57	**100.00±0.00**	**100.00±0.00**
10	81.38±4.38	83.84±0.97	95.81±3.78	84.87±3.15	91.35±1.36	**99.90±0.04**	97.18±0.57	97.50±0.58	94.74±5.05	93.97±1.48	96.08±0.06	97.91±0.15
11	94.08±1.59	97.24±0.28	99.59±0.24	93.68±0.31	98.19±1.48	99.43±0.30	93.79±0.36	98.78±1.54	**100.00±0.00**	98.84±0.67	97.26±0.02	**100.00±0.00**
12	98.78±0.79	96.79±0.04	99.19±0.02	99.83±1.68	99.87±0.12	99.88±0.03	81.88±2.45	98.86±0.24	99.42±0.05	81.08±1.03	**100.00±0.00**	**100.00±0.00**
13	96.43±0.39	97.22±0.14	98.09±1.33	97.45±1.52	94.00±2.94	98.83±0.39	**100.00±0.00**	99.81±0.05	99.86±0.16	99.44±0.89	98.46±1.07	99.54±0.31
14	98.00±0.54	95.53±1.55	99.06±0.38	95.81±0.68	99.31±0.67	97.83±1.10	96.05±0.37	98.72±1.24	98.65±0.99	98.85±0.72	**99.81±0.03**	99.71±0.16
15	75.58±0.45	83.19±3.13	79.81±1.77	97.78±2.98	92.86±3.13	88.79±3.41	86.52±1.46	90.12±3.25	87.07±3.35	79.72±1.34	**97.41±0.03**	97.23±0.21
16	95.99±0.37	92.01±1.15	98.45±1.02	99.59±0.34	99.28±0.58	99.61±0.11	98.88±0.39	95.94±0.78	98.71±1.21	78.90±0.88	98.14±0.03	**99.94±0.06**
OA (%)	85.18±0.45	88.26±1.39	94.67±4.34	91.13±1.89	95.33±2.11	95.79±0.53	95.17±0.06	95.13±1.24	94.32±0.59	86.86±2.81	97.81±0.23	**99.20±0.17**
AA (%)	91.81±1.07	92.65±0.86	97.06±2.29	95.26±1.58	97.15±1.05	97.77±0.37	94.77±0.62	97.39±0.77	97.21±0.29	87.01±2.67	98.82±0.10	**99.53±0.03**
Kappa	83.49±0.65	86.93±0.86	94.07±2.47	90.04±1.67	94.80±1.55	95.31±0.59	94.62±0.07	94.60±1.56	93.88±0.68	85.46±2.74	97.57±0.24	**99.11±0.18**

TABLE IV

COMPARISON ON THE HONGHU DATASET. THE BEST RESULTS ARE IN BOLD

Class	SVM	3D-CNN	SSRN	SF	MGHT	CEGCN	AMGCFN	WFCG	GTFN	S^{2}Mamba	HSI-MFormer	MCF-SSGN
1	94.09±4.99	92.44±2.64	93.12±2.98	93.51±1.58	92.43±2.52	93.49±11.69	93.19±1.86	89.77±7.98	93.18±1.41	95.75±0.92	95.33±0.30	**97.78±0.09**
2	87.04±7.14	91.55±1.86	85.04±1.58	90.30±1.46	95.73±1.44	87.29±9.80	94.42±1.44	92.49±6.53	91.85±2.48	**96.38±0.85**	94.97±0.14	95.20±0.24
3	93.99±6.13	83.80±2.54	96.47±1.43	87.44±2.65	83.33±1.42	**99.41±0.53**	81.04±5.40	89.96±4.82	82.21±6.45	81.74±1.27	85.38±0.17	94.98±0.48
4	74.10±5.97	97.71±1.67	91.28±1.19	86.71±2.74	98.25±0.39	95.95±2.90	**99.60±0.09**	95.14±2.30	99.41±0.32	92.51±1.05	93.19±1.21	98.24±0.04
5	80.11±5.56	88.19±1.95	86.18±2.43	97.77±1.08	92.07±2.54	98.10±1.20	93.70±9.16	96.56±2.62	96.45±1.96	96.51±1.02	97.23±0.62	**98.11±0.33**
6	76.06±2.55	78.02±5.51	91.63±1.80	96.01±1.55	91.08±2.15	95.98±3.48	91.99±4.17	98.06±1.52	92.84±1.34	90.68±0.73	93.69±1.16	**97.36±0.59**
7	91.71±5.31	**98.80±0.73**	83.59±7.64	61.26±6.28	89.83±9.83	78.21±8.83	97.43±1.28	91.66±5.75	92.95±4.36	80.12±1.12	85.64±0.85	85.24±0.77
8	72.41±6.14	74.96±6.79	63.38±9.82	88.45±2.29	83.22±1.54	86.16±14.91	78.19±11.37	84.21±9.79	75.73±6.55	60.31±0.98	90.86±1.12	**96.32±0.31**
9	**98.94±0.02**	91.78±1.40	92.53±2.49	98.85±1.14	93.64±1.62	91.85±4.59	97.37±1.02	92.53±2.63	82.83±1.31	99.00±0.49	97.25±0.45	97.28±0.58
10	62.29±7.60	70.79±3.71	86.16±4.93	97.89±0.95	95.41±1.90	87.27±1.51	90.84±3.98	86.16±4.93	**95.59±1.49**	92.20±1.08	93.61±1.45	93.51±0.89
11	74.61±7.50	70.76±11.32	72.67±9.84	82.27±3.19	**93.40±9.65**	56.43±9.82	79.01±7.90	85.67±4.52	67.70±8.74	76.76±1.41	86.09±0.52	87.95±1.95
12	76.81±6.34	83.89±9.05	80.70±10.68	66.14±6.71	90.12±7.17	89.97±8.77	82.54±7.30	86.70±8.64	79.44±9.64	74.11±1.24	87.79±1.38	**92.08±2.08**
13	70.93±5.06	67.63±8.65	86.48±8.10	71.57±8.49	86.89±0.35	86.11±9.87	**87.89±0.91**	79.48±6.86	69.57±9.56	76.70±0.92	78.28±0.12	87.60±0.24
14	87.93±2.16	66.82±6.12	89.10±5.94	91.99±3.59	83.00±1.61	87.76±9.93	91.74±1.49	82.10±7.61	83.36±5.21	90.02±0.78	93.19±1.13	**97.03±0.73**
15	77.52±6.09	98.44±0.13	83.77±4.30	96.45±1.34	97.08±1.66	98.45±0.64	99.00±0.31	99.77±0.16	99.56±0.42	**100.00±0.00**	**100.00±0.30**	98.07±0.12
16	92.31±2.06	93.89±1.76	93.19±2.86	94.08±2.65	95.64±2.86	93.27±5.49	88.24±0.79	93.19±4.53	87.65±5.72	90.96±0.67	98.05±0.82	**98.29±1.15**
17	82.90±5.53	83.08±5.45	97.02±0.52	89.71±3.46	92.59±2.59	94.49±1.27	92.19±4.25	91.02±8.28	92.13±0.42	96.61±0.84	**96.98±0.96**	93.89±1.88
18	96.25±1.35	89.79±5.03	82.21±1.53	97.08±0.41	96.52±1.88	98.35±1.01	94.19±2.65	92.21±4.30	**99.11±0.21**	96.52±0.83	97.75±0.54	94.33±0.47
19	74.55±6.04	82.20±6.73	87.45±5.36	96.15±1.95	87.04±2.83	93.15±1.74	91.23±1.60	94.04±2.83	**97.04±0.58**	94.51±0.71	93.05±1.41	93.33±0.62
20	79.58±5.29	95.62±1.28	94.68±1.48	97.94±0.05	97.08±3.66	94.62±3.36	98.75±1.08	94.68±11.85	**99.03±0.53**	97.08±0.66	97.83±1.26	98.23±0.59
21	80.52±8.62	92.01±1.19	84.08±5.55	98.92±0.03	97.73±2.43	86.72±2.63	93.47±4.13	91.19±3.67	95.07±2.38	**100.00±0.00**	99.61±0.17	99.69±0.43
22	94.12±1.09	93.57±2.03	92.75±2.75	99.89±0.49	93.46±1.49	98.05±0.10	96.77±2.81	94.75±3.80	89.94±0.05	99.51±0.32	96.48±0.82	**99.65±0.55**
OA (%)	78.83±3.40	87.69±4.56	89.47±2.18	89.13±3.46	94.82±0.59	92.30±1.37	94.22±1.54	92.54±1.34	92.21±1.26	89.37±0.53	91.67±0.35	**95.65±0.17**
AA (%)	82.67±3.09	83.44±4.13	86.98±2.02	91.94±2.58	93.62±1.97	90.50±1.14	91.49±1.27	90.97±1.73	89.07±0.57	89.91±0.61	93.28±0.15	**95.14±0.03**
Kappa	73.36±4.69	84.46±5.67	86.37±2.73	88.22±3.69	93.52±2.37	90.39±2.60	92.73±2.65	90.67±2.73	90.23±4.82	86.79±0.57	89.63±0.42	**94.53±0.18**

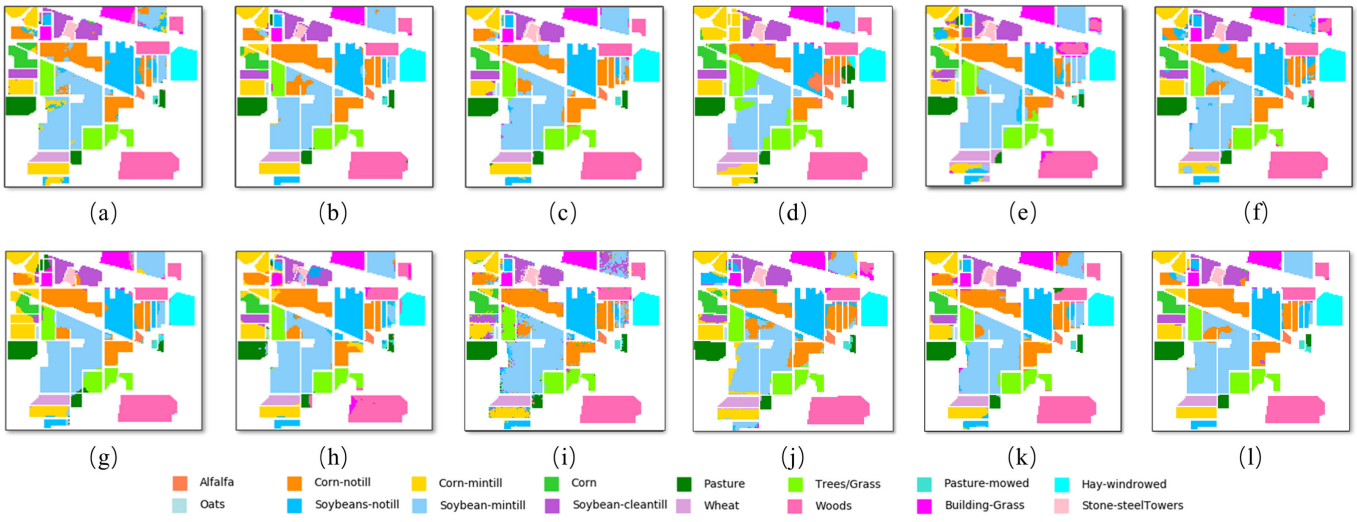


Fig. 7. Visual classification maps on the IP dataset. (a) SVM (73.97%). (b) 3D-CNN (76.25%). (c) SSRN (87.53%). (d) SF (81.54%). (e) MGHT (91.32%). (f) CEGCN (92.51%). (g) AMGCFN (93.40%). (h) WFCG (91.74%). (i) GTFN (90.09%). (j) S^{2}Mamba (86.48%). (k) HSI-MFormer (93.33%). (l) **MCF-SSGN** (95.53%).

fine-grained targets like Classes 6 and 14. These results validate the model's capacity to integrate multiscale spatial–spectral representations and maintain strong generalization across diverse landscapes and category distributions.

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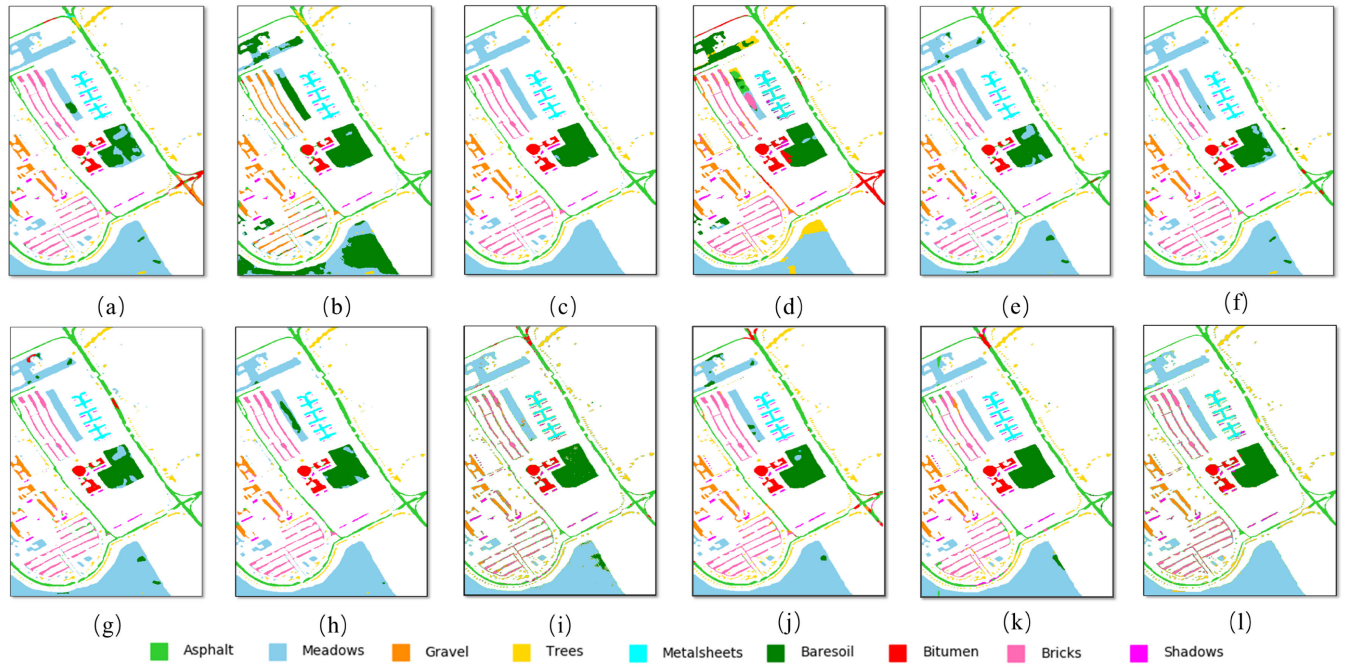


Fig. 8. Visual classification maps on the PU dataset. (a) SVM (83.73%). (b) 3D-CNN (82.29%). (c) SSRN (93.05%). (d) SF (78.17%). (e) MGHT (93.40%). (f) CEGCN (95.40%). (g) AMGCFN (94.83%). (h) WFCG (94.97%). (i) GTFN (91.50%). (j) S^2Mamba (95.12%). (k) HSI-MFormer (94.94%). (l) **MCF-SSGN** (99.26%).

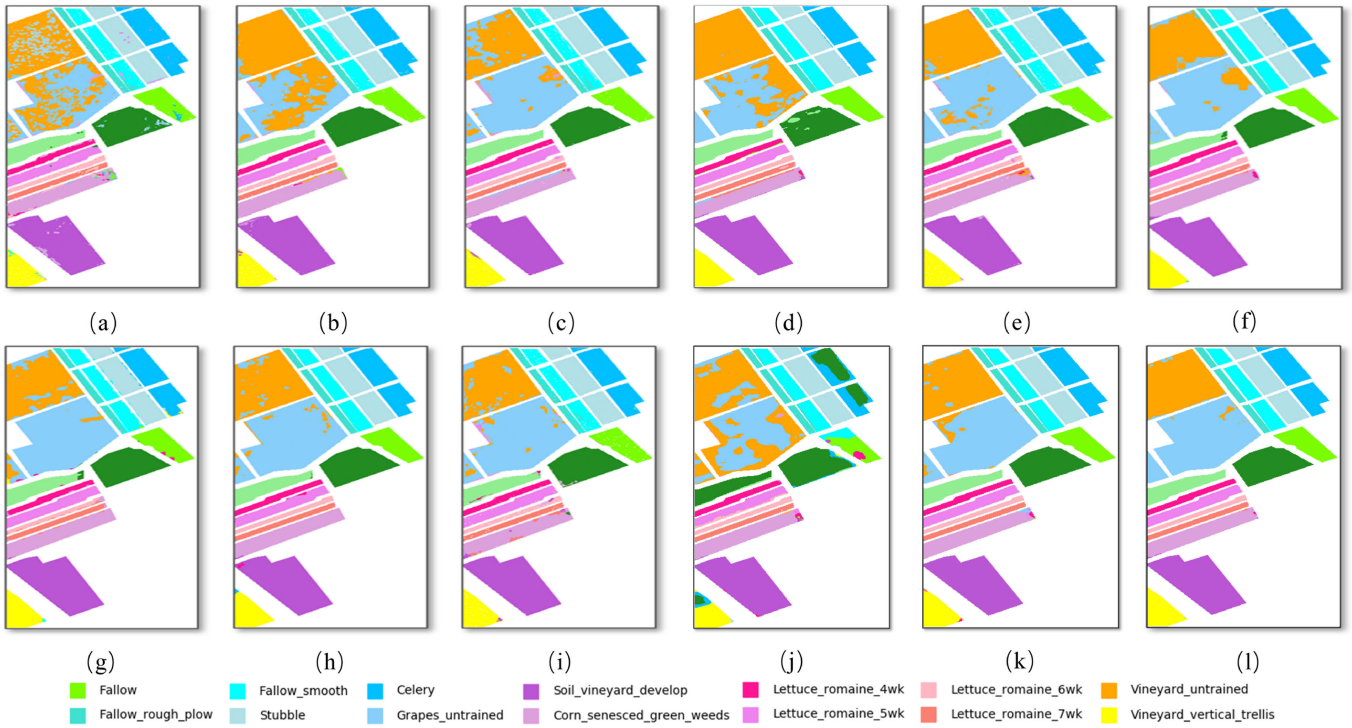


Fig. 9. Visual classification maps on the SA dataset. (a) SVM (85.18%). (b) 3D-CNN (88.26%). (c) SSRN (94.67%). (d) SF (91.13%). (e) MGHT (95.33%). (f) CEGCN (95.79%). (g) AMGCFN (95.17%). (h) WFCG (95.13%). (i) GTFN (94.32%). (j) S^2Mamba (86.86%). (k) HSI-MFormer (97.81%). (l) **MCF-SSGN** (99.20%).

2) Visual Analysis of Classification Results: To further verify the robustness of MCF-SSGN, classification maps on four benchmark datasets—IP, PU, SA, and HongHu—are visualized in Figs. 7–10. Across all scenes, MCF-SSGN yields smoother predictions within homogeneous regions and significantly improves boundary delineation, particularly in fragmented or class-overlapping areas. On IP, MCF-SSGN effectively suppresses speckle noise and enhances structure continuity, especially in Classes 2, 10, and 15, where small. For PU, where urban classes are tightly distributed with sim-

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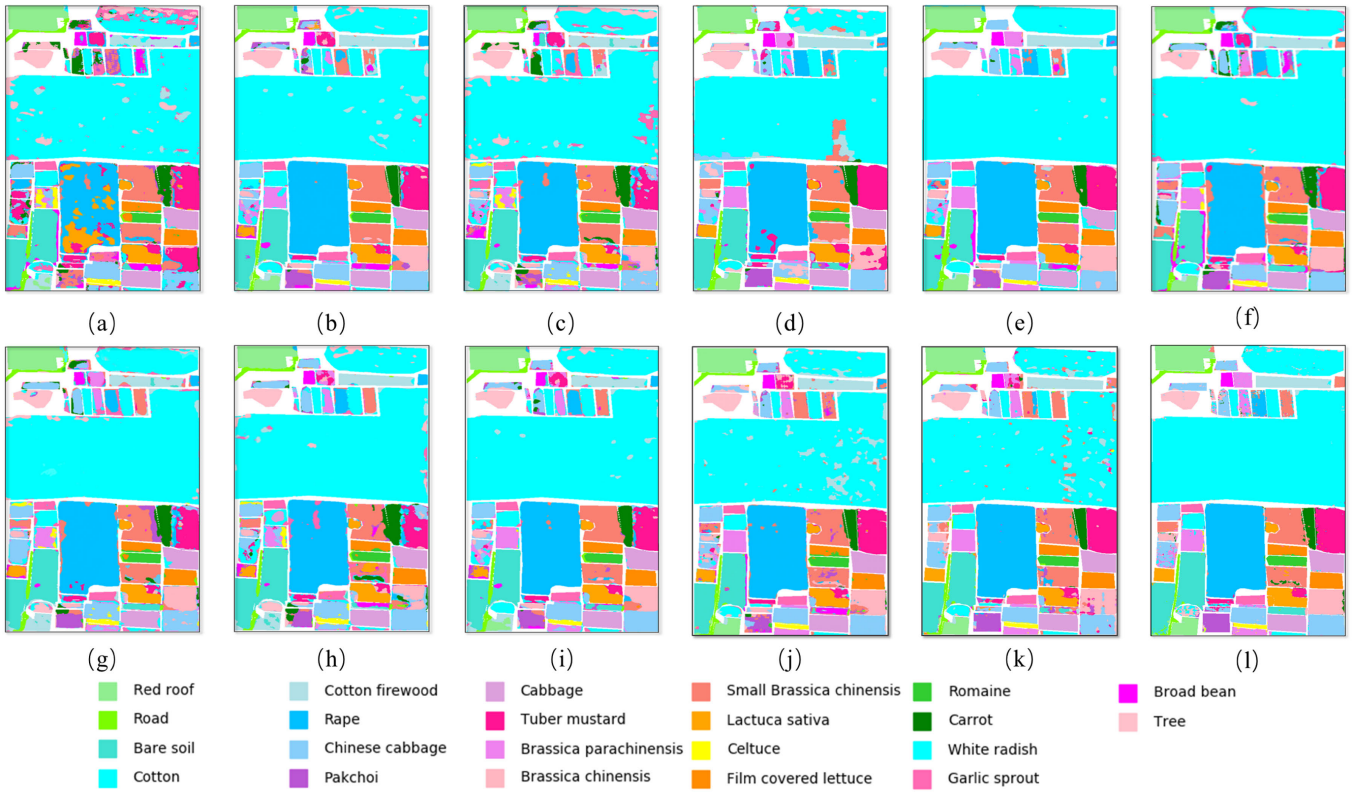


Fig. 10. Visual classification maps on the HongHu dataset. (a) SVM (78.83%). (b) 3D-CNN (87.69%). (c) SSRN (89.47%). (d) SF (89.13%). (e) MGHT (94.82%). (f) CEGCN (92.30%). (g) AMGCFN (94.22%). (h) WFCG (92.54%). (i) GTFN (92.21%). (j) S^{2}Mamba (89.37%). (k) HSIMFormer (91.67%). (l) **MCF-SSGN** (95.65%)

ilar spectra, the proposed method successfully differentiates between Classes 3 and 9, reducing class confusion visible in SSRN and SF predictions. On SA, most methods struggle with spatial coherence in large, regular fields. MCF-SSGN generates clearly segmented boundaries with minimal intraclass noise. For the HongHu dataset, known for its fine-grained and dense vegetation labels, the proposed network shows superior adaptability. Notably, Classes 8 and 13 are better preserved compared to GTFN and WFCG. These visual results support the numerical findings and illustrate MCF-SSGN's ability to jointly capture texture patterns and semantic structure, leading to cleaner, more reliable classification outcomes across diverse landscape scenarios.

V. Discussion

A. Ablation Studies

To evaluate the contribution of each core component, ablation experiments were conducted on four benchmark datasets, focusing on the SA-SGCN, MEAM, and BAFM. The results are presented in Table V.

TABLE V
Ablation Results of Core Modules on Four Datasets

Date	SA-SGCN	MEAM	BAFM	OA (%)	AA (%)	κ
IP	×	√	×	90.25	94.74	88.88
	√	×	×	83.16	87.04	80.54
	√	√	×	92.08	95.57	90.98
	√	√	√	**95.21**	**97.44**	**94.52**
PU	×	√	×	94.78	96.50	93.16
	√	×	×	87.18	86.37	83.35
	√	√	×	98.70	99.10	98.29
	√	√	√	**99.34**	**99.06**	**99.12**
SA	×	√	×	94.10	97.15	94.20
	√	×	×	98.52	97.42	98.35
	√	√	×	98.55	97.53	98.50
	√	√	√	**99.10**	**99.51**	**99.00**
HongHu	×	√	×	78.68	74.23	73.80
	√	×	×	94.09	93.52	92.58
	√	√	×	93.37	93.32	91.68
	√	√	√	**95.78**	**95.38**	**94.69**

Notes: "√" indicates that the module is enabled; "×" indicates that the module is disabled. Each configuration was run ten times, and the mean value is reported. Best results are shown in bold.

The complete MCF-SSGN consistently delivered the highest accuracy across all datasets. On IP, the OA reached 95.21%, representing a 3.13% improvement over the variant without BAFM, underscoring the importance of adaptive cross-branch fusion in cropland scenes. On PU, the OA increased from 98.70% to 99.34% when BAFM was added, confirming the complementary effect among the three modules in urban environments. On HongHu, BAFM improved OA from 93.37% to 95.78%, further validating its ability to integrate spatial and structural cues in farmland-dominated scenarios. In isolation, SA-SGCN showed notable standalone performance, achieving 98.52% OA on SA, reflecting its effectiveness in capturing spectral distinctions through adaptive topology. However, on datasets like IP and HongHu, the integration via BAFM provided larger gains by balancing local CNN features and global graph representations. Overall, SA-SGCN enables structure-

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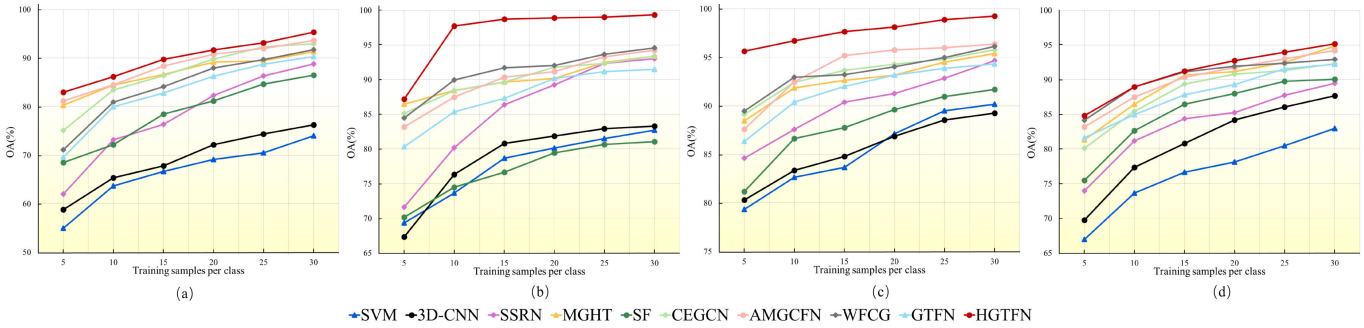


Fig. 11. OA versus the number of training samples per class on four datasets. (a)–(d) IP, PU, SA, and HongHu datasets, respectively.

TABLE VI
IMPACT OF DIFFERENT INTEGRATION MODULES

OA	Add	Concatenation	AFF	SENet	DGF	BAFM
IP	91.58	93.28	95.18	94.01	94.71	**95.53**
PU	90.03	92.05	97.13	95.17	97.58	**99.26**
SA	92.21	93.82	96.97	96.04	96.13	**99.20**

aware reasoning, MEAM enhances spectral–spatial features at multiple scales, and BAFM adaptively fuses information across branches. Their synergy accounts for the robust and superior performance of MCF-SSGN.

B. Ablation Experiments on the Fusion Module

To demonstrate the effectiveness of the BAFM, we compared it with five fusion strategies: element-wise Add, Concatenation, and three advanced adaptive modules (attentional feature fusion (AFF) [46], [47], squeeze-and-excitation (SENet) [48], and dynamic gated fusion (DGF) [49]). As shown in Table VI, static methods (Add and Concatenation) yield suboptimal results due to severe feature interference between heterogeneous CNN and GCN representations. While adaptive modules (AFF, SENet, and DGF) improve performance by dynamically recalibrating feature weights, they operate exclusively at the feature level. In contrast, our BAFM consistently achieves the highest OA across all datasets (e.g., 99.26% on PU). By employing a highly nonlinear, dual-level fusion scheme, BAFM effectively prevents linear degradation and maximizes the complementary potential of cross-domain features.

C. Model Complexity and Efficiency

In addition to classification accuracy, model complexity and efficiency are essential considerations for real-world deployment. Table VII presents a comparison of training/testing times and parameter sizes across four hyperspectral datasets.

Traditional convolution-based models (e.g., 3D-CNN, HybridSN, and SSRN) rely on deep stacking, leading to high parameter counts (all exceeding 490k) and prolonged training times, particularly on HongHu, where training exceeds 150 s. Transformer-based methods (e.g., WFCG and GTFN) attain strong accuracy but incur substantial costs—WFCG, for instance, requires over 2200 s training time and 1.1 M parameters on HongHu, posing challenges for latency-sensitive scenarios. Graph-based baselines (e.g., CEGCN and AMGCFN) offer lower parameter counts, yet their training and inference speeds remain suboptimal, especially in large-scale or complex scenes. In contrast, the proposed MCF-SSGN achieves a better tradeoff between accuracy and efficiency. With only 337.62k parameters, it is significantly lighter than most convolutional and Transformer counterparts. Training completes within 100 s on HongHu, and testing on PU takes just 3.49 s—about 17% and 73% faster than HybridSN and GTFN, respectively. Notably, MCF-SSGN trains on IP in just 6.34 s, outperforming all competitors by a substantial margin.

D. Generalization Performance Analysis

To evaluate generalization under limited supervision, we varied the number of labeled samples per class from 5 to 30 on four datasets. As shown in Fig. 11, MCF-SSGN consistently achieves the highest OA across all sample sizes. Under five-shot settings, it outperforms classical baselines (SVM and 3D-CNN) by over 15% and even surpasses competitive models like WFCG and AMGCFN by 2%–4%. On IP, the model surpasses 90% OA at just 15 samples/class, while GTFN requires 30 to reach similar performance. On PU, MCF-SSGN reaches 98% OA at 10 samples/class and remains nearly saturated thereafter. On SA, a steady increase is observed, with 98% OA achieved at 20 samples/class—1.2% higher than AMGCFN. On HongHu, MCF-SSGN maintains a clear lead beyond 20 samples/class. These results highlight the model's strong generalization, fast convergence, and robustness in diverse scenes, benefiting from its dual-branch fusion and spectral–spatial co-modeling.

E. Full-Scene Visualization and Spatial Consistency Analysis

To further evaluate the spatial robustness and generalization capability of MCF-SSGN, we provide full-scene classification maps for all four benchmark datasets, as illustrated in Fig. 12. Unlike standard classification maps that mask out unlabeled pixels, full-scene visualization requires the model to assign a semantic label to every pixel in the HSI cube, including the background and transitional areas.

The experimental results demonstrate that MCF-SSGN exhibits exceptional spatial consistency across diverse landscapes. In the Salinas and PU scenes, the predicted background

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TABLE VII
COMPARISON OF THE OPERATIONAL EFFICIENCY OF DIFFERENT METHODS

Method	IP		PU		SA		HONGHU		Parameters
	Train (s)	Test (s)	Train (s)	Test (s)	Train (s)	Test (s)	Train (s)	Test (s)	HongHu ($\times 10^3$)
3D-CNN	87.96	4.82	54.87	10.34	93.43	11.45	139.16	116.51	585.57
HybridSN	109.08	3.24	60.81	11.49	115.04	13.56	206.16	230.15	493.36
SSRN	105.24	3.39	78.12	10.55	117.72	11.98	154.97	225.26	722.40
MGHT	145.72	2.75	183.32	13.98	117.47	10.83	295.48	34.61	965.76
CEGCN	45.75	2.35	89.46	10.21	53.27	6.85	216.97	35.12	176.24
AMGCFN	86.23	1.14	199.93	14.15	156.25	9.15	574.35	59.92	386.70
WFCG	282.34	3.98	589.12	16.45	729.55	11.45	2264.77	68.28	1119.12
GTFN	121.42	18.35	297.65	49.56	310.13	52.56	1947.97	563.54	282.66
S^2Mamba	145.32	9.14	169.83	24.50	319.84	42.15	1498.76	308.16	368.04
HSI-MFormer	124.17	2.42	149.65	8.96	144.40	11.98	156.86	149.60	1749.22
MCF-SSGN	**6.34**	**0.35**	**38.61**	**3.49**	**24.72**	**1.79**	**88.67**	**8.61**	**337.62**

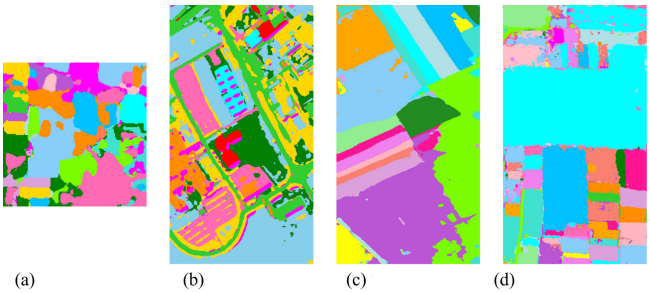


Fig. 12. Full-scene classification maps on four benchmark datasets. (a) IP (95.17%). (b) PU (99.23%). (c) SA (99.06%). (d) HongHu (95.20%).

regions are characterized by homogeneous color blocks and sharp class boundaries, indicating that the model effectively mitigates the "salt-and-pepper" effect common in pixel-wise classifiers. For the more fragmented IP and WHU-Hi-HongHu datasets, the model maintains high structural continuity even in narrow corridors and overlapping field boundaries. This superior performance is primarily due to the SA-SGCN branch, which leverages superpixel-level structural priors to enforce regional smoothness, and the MEAM branch, which adaptively captures multiscale textures to preserve edge details. Such holistic visualizations confirm that MCF-SSGN is not only accurate within the labeled test sets but also yields reliable and interpretable classification outcomes across the entire spectral–spatial domain, enhancing its practical utility for large-scale environmental monitoring.

VI. CONCLUSION

GCNs capture non-Euclidean spectral–spatial relations, while CNNs extract structured Euclidean features. To exploit their complementarity, we propose MCF-SSGN, a MCF-SSGN for fast hyperspectral classification. A MEAM adaptively generates kernels for spatial flexibility and reduced cost, and an SA superpixel graph convolution enhances informative band aggregation. A dual-branch adaptive fusion balances CNN and GCN representations beyond static fusion. On four benchmark datasets (IP, PU, SA, and HongHu), MCF-SSGN achieves OA of 95.53%, 99.26%, 99.20%, and 95.65% with per-sample inference times of 0.35–8.61 s, all under 10 s. These results demonstrate robust and efficient spectral–spatial complementary fusion for HSIC.

In future research, we will focus on developing unsupervised and semi-supervised extensions to reduce reliance on labeled data [50], [51] while enhancing the framework's generalizability across diverse scenes and sensors [52]. Furthermore, inspired by the progressive hybrid-order hypergraph paradigm [53], we aim to refine the graph construction process by incorporating higher order nonlinear relationships and graph-hypergraph collaborative learning, thereby capturing more complex multiscale spectral–spatial dependencies and further improving the model's robustness for large-scale and heterogeneous remote sensing applications.

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